SYMPLECTIC GEOMETRY

github.com/danimalabares/stack

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See github.com/danimalabares/sg for the complete notes of Henrique Bursztyn's course.

1. Symplectic vector spaces

Proposition 1.1. *Every symplectic vector space (V, ω) admits a symplectic basis, i.e. a basis $e_1, \ldots, e_n, f_1, \ldots, f_n$ such that*

$$\omega(e_i, e_j) = 0, \qquad \omega(e_i, f_j) = 1, \qquad \omega(f_i, f_j) = 0.$$

Proof. Like Gram-Schmidt, start with any vector e_1, then there exists another vector f_1 such that $\omega(e_1, f_1) = 1$. The space generated by e_1, f_1 is symplectic, so we get a normal decomposition... □

Exercise 1.2. Let V be a vector space and ω a 2-form on V. Show that ω is nondegenerate if and only if ω^n is not zero.

Proof. If ω is nondegenerate, by Proposition 1.1 there is a symplectic basis $e_1, \ldots, e_n, f_1, \ldots, f_n$ such that $\omega = \sum e^i \wedge f^i$. Then $\omega^n = n! e^1 \wedge \ldots \wedge e^n \wedge f^1 \wedge \ldots \wedge f^n$. Hence evaluating in $e_1, \ldots, e_n, f_1, \ldots, f_n$ we see $\omega^n \neq 0$.

For the converse we suppose that ω is degenerate and prove that $\omega^n = 0$. It follows from the following property of interior product:

$$i_v \omega^n = n i_v \omega \wedge \omega^{n-1} \qquad \forall v \in V.$$

Supposing that ω is degenerate we have $v \in V$ such that $i_v \omega = 0$, which gives $i_v \omega^n = 0$. Completing v to a basis $v_2, \ldots, v_n$ we see that $\omega^n(v, v_2, \ldots, v_n) = i_v \omega^n(v_2, \ldots, v_n) = 0$. □

2. Moser's trick

Theorem 2.1 (Moser's trick)**.** *Let M be compact and $\{\omega_t\}$ symplectic forms with $t \in [0,1]$. If all these forms are mutually cohomologous, that is $[\omega_t]$ is constant. Then there exists ϕ_t such that $\phi^*\omega_t = \omega_0$ for all t.*

3. Weinstein's Lagrangian submanifold theorem

It was Weinstein who said that everything is a Lagrangian submanifold. The following theorem says that Lagrangian submanifolds are defined intrinsically in the sense that they can all be thought of as embedded in its own cotangent bundle.

Theorem 3.1 (Weinstein's Lagrangian submanifold theorem)**.** *If $L \subset M$ is a Lagrangian submanifold, there is are neighbourhoods $L \subseteq U_0 \subset M$ and $L \subset U_1 \subset T^*L$ and a symplectomorphism $\varphi : U_0 \to U_1$.*

4. Coadjoint orbits and KKS theorem

The coadjoint action induces orbits in $\mathfrak{g}^*$. The tangent space at any point of a coadjoint orbit is given by infinitesimal vector fields, that is,

$$T_\xi \mathcal{O} = \{u_{\mathfrak{g}^*}(\xi) : \forall u \in \mathfrak{g}\}.$$

We may introduce a symlpectic form $\omega \in \Lambda^2 T^*_\xi \mathcal{O}$ on the tangent space at some point of a coadjoint orbit by

$$\omega_\xi(X,Y) := \xi([u,v]) = -u_{\mathfrak{g}^*}(\xi)v = v_{\mathfrak{g}^*}(\xi)(u)$$

which does not depend on $u, v \in \mathfrak{g}$.

Theorem 4.1 (KKS)**.** *$\xi \mapsto \omega_\xi$ is symplectic on $\mathcal{O}$.*

Proof. Non degeneracy is easy. Then show that ω is G-invariant. Closedness follows from Jacobi identity. □

5. Hamiltonian actions and moment map

To warm-up we define real symplectic and Hamiltonian actions.

Definition 5.1.
- An $\mathbb{R}$-action is *symplectic* if $\psi_t^* \iff L_X\omega = 0$, that is, X is a symplectic field.
- An $\mathbb{R}$-action is *hamiltonian* if X is Hamiltonian, that is, if there exists a function $H \in C^\infty(M)$ such that $X = X_H$.

To define moment map suppose there exists a map $\hat{\mu}$ such that

$$\begin{tikzcd}
 & C^\infty(M) \arrow[d] & f \arrow[d, mapsto] \\
\mathfrak{g} \arrow[ur, "\hat{\mu}"] \arrow[dr] & \mathfrak{X}(M) & X_f \\
u \arrow[r, mapsto] & u_M \arrow[ur, equal] &
\end{tikzcd}$$

Note that if $\hat{\mu}$ exists, it's not uniquely determined, and that we may suppose that it is linear.

It's equivalente to define $\hat{\mu}$ as a map $\hat{\mu} : \mathfrak{g} \to C^\infty(M)$ or as a map

$$\begin{aligned} \mu : M &\longrightarrow \mathfrak{g}^* \\ \hat{\mu}(u)(x) &= \langle \mu(x), u \rangle . \end{aligned}$$

Definition 5.2.

- A G-action is symplectic if $\psi_g^* \omega = \omega$ for all $g \in G$, that is, if

$$\begin{tikzcd} \psi : G \arrow[r] \arrow[dr] & \mathrm{Dif}(M) \\ & \mathrm{Symp}(M, \omega) \arrow[u, hook] \end{tikzcd}$$

- A symplectic action $G \overset{\psi}{\curvearrowright} (M, \omega)$ is *weakly Hamiltonian* if there exists a map $\mu : M \to \mathfrak{g}^*$ (or equivalently, $\hat{\mu} : \mathfrak{g} \to C^\infty(M)$ linear) such that

$$i_{u_M} \omega = d \langle \mu, u \rangle , \qquad (\text{resp. } u_M = X_{\hat{\mu}(u)}).$$

- A weakly Hamiltonian action is *Hamiltonian* if it is also μ-equivariant, that is, $\mu : G \curvearrowright M \to g^* \overset{\mathrm{Ad}^*}{\curvearrowleft} G$,

$$\mu \circ \psi_g = (\mathrm{Ad}^*)_g(\mu).$$

In this case we call μ a *moment map*.

6. Noether's principle

Theorem 6.1 (Noether's principle)**.** *Let (M, ω, μ) be a Hamiltonian G-space with Hamiltonian $H \in C^\infty(M)$. H is G-invariant if and only if μ is preserved by the Hamiltonian flow.*

Proof. one line □

7. Symplectic quotient

In symplectic geometry we cannot always take quotients. We need to use level-sets of the moment map.

Theorem 7.1 (Marsden-Weinstein, Megre)**.** *Let $G \curvearrowright (M, \omega)$ be a Hamiltonian action with moment map $\mu : M \to \mathfrak{g}^*$. Suppose $0 \in \mathfrak{g}^*$ is a regular value. By the equivariance of μ, we have an action $G \curvearrowright \mu^{-1}(0)$ given as follows. For $x \in \mu^{-1}(0)$, $\mu(\psi_g(x)) = Ad_g^*(\mu(x)) = 0$. Suppose that this action is regular (this will be defined later), that $\mu^{-1}(0)/G$ is smooth, and that the projection is a submersion. Then there exists a unique symplectic form ω_{red} in M_0 such that $i^*\omega = \pi_0^* \omega_{red}$.*

That is, the pullback of the symplectic form to the 0 level-set of μ passes to the quotient:

$$\begin{tikzcd} G \curvearrowright \mu^{-1}(0) \arrow[r, hook] \arrow[d, "\pi_0"] & M \\ (M_0 := \mu^{-1}(0)/G, \omega_{\mathrm{red}}) & \end{tikzcd}$$

A slightly more general setting is the following. Hamiltonian action $G \curvearrowright (M, \omega)$, and moment map $\mu : M \to \mathfrak{g}^*$. $\xi \in \mathfrak{g}^*$ regular, $G_\xi \subseteq G$ stabilizer of ξ. Then we have

$$\begin{tikzcd} G_\xi \curvearrowright \mu^{-1}(\xi) \arrow[r, hook, "i_\xi"] \arrow[d, "\pi_\xi"] & M \\ (M_\xi = \mu^{-1}(\xi)/G_\xi, \omega_{\mathrm{red}}). \end{tikzcd}$$

8. Toric varieties

For any compact connected symplectic manifold with an action of a torus, $\mathbb{T}^n \curvearrowright (M^{2n}, \omega) \xrightarrow{\mu} \mathbb{R}^n = (\mathfrak{t}^n)^*$, the image of μ is a polytope. By [Guilleimin-Sternberg] and [Atiyah], this polytope is convex. Further, if the dimension of the torus is exactly half the dimension of the manifold, by [Delzant], the moment polytope determines the manifold, i.e. there is a one-to-one correspondence between Delzant polytopes and symplectic toric varieties. Moduli space of polygons is an example of this, see [KM95].

In fact, this situation of having a torus action of half the dimension is what we may call a "totally integrable system".

9. (Completely) integrable systems

Let (M, ω) be a symplectic manifold. Recall that

- $\{f, g\} = \omega(X_g, X_f) = L_{X_f} g = -L_{X_g} f$.
- $X_{\{f,g\}} = [X_f, X_g]$

If $\{f, g\} = 0$ then the Hamiltonian flows of f and g commute.

Definition 9.1. Let (M^{2n}, ω) and $H \in C^\infty(M)$. (M, ω, H) is a *completely integrable system* (in Liouville's sense) if there exist functions $f_1, \ldots, f_n \in C^\infty(M)$ such that

(1) $H = f_1, \ldots, f_n$ are linearly independent ($df_1, \ldots, df_n$ are linearly independent in every point).
(2) $\{f_i, f_j\} = 0$ for all i, j.

Note that $F = (f_1, \ldots, f_n) : M \to \mathbb{R}^n$ is a submersion. For all $c \in \mathbb{R}^n$, $F^{-1}(c)$ are Lagrangian submanifolds. $F^{-1}(c)$ is invariant by the flow of X_{f_i} for $i = 1, \ldots, n$.

Explanation: $F^{-1}(c)$ is a level-set submanifold of M, whose tangent space is generated by the df_i. These are all in the kernel of dF by the integrability condition, namely $dF = (df_1, \ldots, df_n)$, $df_i(X_{f_j}) = \{f_i, f_j\} = 0$.

The idea is that finding these "conserved quantities" we manage to describe the Hamiltonian dynamics completely. Showing that a system is integrable is a victory. The moduli space of polygons is an example, and I think that so is $(\mathbb{C}^2)^{[n]}$.

There is another method of using "bihamiltonian actions" (two Hamiltonian actions yielding the same dynamics) /Lax pairs (defining the system using eigenvalues of matrices). Examples of this are Toda, Caolengo-Moser.

The following theorem says that the connected component of a symplectic manifold with an integrable system

Theorem 9.2 (Arnold-Liouville)**.** *Let (M^{2n}, ω), $H = f_1, \dots, f_n$ be completely intgrable system and $F : M \to \mathbb{R}^n$, $F = (f_1, \dots, f_n)$. Let M_c be the connected component of $F^{-1}(c)$. Then*

1. *If M_c is compact, then $M_c \simeq \mathbb{T}^n = \mathbb{R}^n/2\pi\mathbb{Z}^n = \{(\theta_1, \dots, \theta_n) \text{ mod } 2\pi\}$.*
2. *In angular coordinates, the Hamiltonian flow is linear, that is, $\frac{d}{dt}^{-1}(t) = \alpha \in \mathbb{R}^n$ implies that $\theta(t) = \theta_0 + \alpha t$.*

The proof of item 1 is given by the following proposition putting $N = M$ and $X_1 = X_{f_1}, \dots, X_n = X_{f_n}$.

Proposition 9.3. *Let N be complete, connected, of dimension n. Let $X_1, \dots, X_n$ be linearly independent vector fields on N such that $[X_i, X_j] = 0$ for all j. Then $N \simeq \mathbb{T}^n$.*

Proof. First we present the steps:

1. Flows of $X_1, \dots, X_n$ define an action $\mathbb{R}^n \curvearrowright N$.
2. This action is transitive. Then $N \simeq \mathbb{R}^n/\Gamma$ where Γ is the stabilizer.
3. Since $\dim N = n$, then Γ is a lattice of $\mathbb{R}^n$.
4. Since N is compact, then Γ is of maximum rank, that is, $\Gamma = 2\pi\mathbb{Z}^n \subset \mathbb{R}^n$.

Let φ_t^i be the flow of X_i (which are complete since N is compact). Since $[X_i, X_j] = 0$, then $\varphi_t^i \varphi_s^j = \varphi_s^j \varphi_t^i$. Thus we have an action $\mathbb{R}^n \curvearrowright N$ given by moving along the flows of the vector fields, that is

$$\underbrace{(t_1, \dots, t_n)}_{\mathbf{t}} \cdot x = \underbrace{\varphi_{t_1}^1 \circ \dots \circ \varphi_{t_n}^n(x)}_{\varphi_{\mathbf{t}}(x)}$$

To prove transitivity we can show that the orbit maps of this action are surjective. Indeed, the image of the orbit map is open since it is a local diffeomorphism (it's derivative is given by the vector fields, which are linearly independent!), and it is closed by a taking a point in the boundary and using the flow at this point (and commutativity of the flows).

To check that the stabilizer Γ of this action is trivial first note that all stabilizers are equal (not only conjugate) since $\mathbb{R}^n$ is abelian. Further, since the orbit map is a local diffeomorphism, Γ is discrete. Thus, it must be a lattice of rank $k \leq n$.

To show that in fact $k = n$ consider a linear map T that maps the generators of Γ to the first k canonical vectors of $\mathbb{R}^n$. By the following diagram, the arrow on the bottom is well defined as in fact a bijective local diffeomorphism, that is, a diffeomorphism.

$$\begin{tikzcd}
\mathbb{R}^k \times \mathbb{R}^{n-k} \simeq \mathbb{R}^n \arrow[r, "T"] \arrow[d, "p"'] \arrow[dr, "\psi_x \circ T"] & \mathbb{R}^n \arrow[d, "\psi_x = \text{orbit map}"] \\
\mathbb{T}^k \times \mathbb{R}^{n-k} \arrow[r, "\overline{T}"'] & N
\end{tikzcd}$$

Thus the factor $\mathbb{R}^{n-k}$ must be a point since N is compact by hypothesis. □

Item 2 from Theorem 9.2 is obtained by considering the flows on the latter diagram and is left as an exercise.

In the case that N is not compact we just conclude that it M_c is a product of a torus with an Euclidean space.

Perhaps the simplest example of an integrable system is given by taking an open ball about the origin $B \subseteq \mathbb{R}^n$ and $B \times \mathbb{T}^n = \{(I_1, \ldots, I_n, \phi_1, \ldots, \phi_n)\}$ with the symplectic form $\omega = \sum_{i=1}^n dI_1 \wedge d\phi_i$ with $\{I_1, I_j\} = 0$.

The following theorem says that in fact the **symplectic form** of every level set of the kind M_c, which we just proved that is a torus, may is described locally as such a product.

Theorem 9.4. *There is a neighbourhood U of M_c and a symplectomorphism*

$$\begin{tikzcd}
\sum d\mathcal{I} \wedge d\phi_i & B \times \mathbb{T}^n \arrow[l, squiggly] \arrow[r, "r", "\simeq"'] \arrow[d, "p"'] & U \subseteq M \arrow[r, squiggly] \arrow[d, "F"] & \omega \\
& B \arrow[r, "\simeq"'] & F(U) &
\end{tikzcd}$$

The coordinates $(I_1, \ldots, I_n, \phi_1, \ldots, \phi_n)$ are called *action-angle* coordinates.

The case when $h(I)$ does not depend on the angle coordinates ϕ_i we can solve the Hamiltonian equations easily:

$$\begin{cases} \dot{I}_i = 0 \\ \dot{\phi}_i = -\frac{\partial h}{\partial I}(I) \end{cases}$$

so that $\phi_i(t) = \phi_0 + \alpha t$ for some constants α. This is why integrable systems are actually easy to solve!

10. Vertex algebras

Mônadas, álgebras sobre mônadas. Exemplo: ultrafiltros (Manes, 1976). Campo quântico, como serie de Fourier. Exemplo: Heisenberg algebra.

Problema: o producto de campos quânticos não é bem definido.

References

[KM95] Michael Kapovich and John Millson, *On the moduli space of polygons in the Euclidean plane*, Journal of Differential Geometry **42** (1995), no. 1, 133 – 164.